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Semantic Modeling of a Histopathology Image Exploration and Analysis Tool

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Abstract of the manuscript

Recently, Anatomic Pathology (AP) has seen the introduction of several tools such as high-resolution histopathological slide scanners, efficient software viewers [1]–[3] for large scale histopathological images and virtual slide technologies. These initiatives created the conditions for a broader adoption of computer-aided diagnosis based on whole slide images (WSI) with the hope of a possible contribution to decreasing inter-observer variability.
Beside this, automatic image analysis algorithms represent a very promising solution to:

- support pathologist's laborious tasks during the diagnosis process
- create a quantification-based second opinion
- enhance inter-observer agreement.

In particular, there have been amazing advances in terms of recognition rate and accuracy by recent developments [1], [4], [5].

Similarly, in order to reduce inter-observer variability between AP reports of malignant tumours, the College of American Pathologists edited 67 organ-specific Cancer Checklists and associated Protocols (CAP-CC&P) [6]. Each checklist includes a set of AP observations that are relevant in the context of a given organ-specific cancer and have to be reported by the pathologist. The associated protocol includes interpretation guidelines for most of the required observations.
All these changes and initiatives bring up a number of scientific challenges such as the sustainable management of the available semantic resources associated to the diagnostic interpretation of AP images by both humans (pathologists) and computers (image analysis algorithms). In this context, reference vocabularies and formalization of the associated knowledge are especially needed to annotate histopathology images with labels complying with semantic standards.

Current terminology systems for AP structured reporting (APSR) gather terms of very different granularity [7], [8] and have not yet been compiled in a systematic approach. Moreover, the Integrating Healthcare Enterprise (IHE) APSR template provides a formal representation of only high-level AP observations resulting from human interpretation of low-level morphological abnormalities. There is still a need to extend the scope of IHE APSR and to integrate in a unique formal representation both high-level AP entities observable by humans and the corresponding low-level morphological abnormalities, especially those that can be quantified using image analysis tools.
In this research work, we present our contribution in this direction. We propose a sustainable way to bridge the content, features, performance and usability gaps [9][10] between histopathology and WSI analysis. Our multi-disciplinary approach covers the histopathology and imaging domains. It is structured as follow:

Histopathology domain:

i. Identify and extract relevant quantifiable observations from the College of American Pathologists (CAP) organ-specific Cancer Checklists and associated Protocols (CC&P)
ii. Identify within the reference biomedical ontologies made accessible by the NCBO Bioportal [11], [12] and within the UMLS metathesaurus [13] the available histopathological formalized knowledge covering the scope of CAP-CC&Ps
iii. Build a sustainable visual representation of this knowledge using the semantic types of the UMLS metathesaurus [14], [15].

iv. Initiate a formal representation of this knowledge under the AP Quantifiable Observation termino-ontology

Imaging domain:

i. Identify effective histopathology imaging methods highlighted by recent Digital Pathology (DP) contests
ii. Identify relevant imaging formalized knowledge within the reference biomedical ontologies in NCBO Bioportal [11], [12] and within the UMLS metathesaurus [13]
iii. Extract the imaging terms and functionalities issued from major biomedical-imaging software (MATLAB, ITK, ImageJ)
iv. Initiate a formal representation by integrating this imaging knowledge (issued from contests and biomedical-imaging software) under the Practical Image Processing Tasks termino-ontology

In both histopathology and imaging approaches, a semi-automatic annotation process was used to label the quantitative parameters and relevant terms with codes from predefined reference semantic resources.

In the histopathology domain, two (2) medical experts independently identified relevant terms corresponding to "gold standard" quantitative parameters observed by pathologists to score or grade malignant tumours. F-measure score were calculated to evaluate concordance between experts.

In the imaging domain, relevant terms and functionalities issued from major biomedical-imaging software were extracted manually. Their hierarchization and integration were respectively performed with Protégé® and the Ontology Alignment Graphical User Interface[16] (OnAGUI).

Based on NCBO Bioportal and UMLS semantic types, the concepts and metadata generated represent a sustainable vocabulary, dedicated to histopathology, being able to effectively support daily work on WSI. Semantic models and reference terminologies are essential in DP, being able to support the reproducibility and quality of the diagnostic, to assist and standardize anatomopathological reporting, and to enable multi-center clinical collaboration or research, especially in the context of cancer grading[7].

This research work is a step forward to organized, cross-disciplinary, information-driven collaborations in the histopathological imaging field. Future works should focus on further development toward realizing our longer term goals of advancing interoperability of histopathological imaging systems and reproducibility of histopathological imaging assays [17], [18].

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